

Is HBase a Helpful Database?



HBase is a non-relational database that runs on top of HDFS. This article gives an overview of HBase, discussing its benefits and limitations.

HBase is an open source, distributed, non-relational database, developed by the Apache Software Foundation, which runs on top of HDFS. Initially, it was referred to as Google Big Table, and later re-named HBase. Mainly written in Java, it is a data model that is designed to provide quick random access to a large amount of structured data.

One can store data in HDFS either directly or through HBase. It is natively integrated with Hadoop and works flawlessly alongside other data access engines through YARN.

HBase's features

1. HBase tables are distributed in the cluster via regions, which are automatically split and redistributed as the data grows.
2. HBase is linearly scalable and has automatic failure support.
3. It integrates with Hadoop, both as a source and as a destination.

4. HBase supports an easy-to-use Java API for programmatic access.

Architecture

HBase's architecture is composed of three types of components — the client library, a master server and a region server, the last of which is optional as it can be used based on requirements.

Master server: This acts as a monitoring agent and monitors all region server instances present in the cluster. It also operates as an interface for all the metadata changes. It maintains the state of the cluster by negotiating the load balancing. It is responsible for schema changes and other metadata operations such as the creation of tables and column families.

Regions: Regions are the basic building elements of the HBase cluster, which consists of the distribution of tables and column families. They contain multiple stores, one for each column family. They comprise mainly two components,